



CS607-ARTIFICIAL INTELLIGENTS

(Solved SUBJECTIVE)

CURRENT PAPER



Junaidfazal08@gmail.com
Bc190202640@vu.edu.pk

FOR MORE VISIT
VULMSHELP.COM

JUNAID MALIK
0304-1659294



AL-JUNAID TECH INSTITUTE

**PAID SERVICE
CS619 PROJECTS**

Available training courses

- HTML
- JQUERY
- PHPMYSQL
- JAVASCRIPT
- BOOTSTRAPS
- NODE.JS
- REACT.JS
- CSS

LMS HANDLING

**PAID
ASSIGNMENTS , QUIZ & GDB**

**95% RESULTS
ALL LMS ACTIVITIES**



Contact Us :

+92 304 1659294

www.vulmshelp.com

junaidfazal08@gmail.com

AL-JUNAID TECH INSTITUTE

1. Three phases of machine learning

ANSWER:

- ❖ TRAINING
- ❖ Validation
- ❖ Application

2. soft computing the most communication

ANSWER:

The most common combinations are of the pairs

- ❖ genetic algorithms – fuzzy systems (genetic fuzzy)
- ❖ Neural Networks – fuzzy systems (neuro-fuzzy systems)
- ❖ Genetic algorithms – Neural Networks (neuro-genetic systems)

3. Disadvantage of artificial neural network (ANN)

ANSWER:

- ❖ Training time and complexity
- ❖ Overfitting
- ❖ Interpretability

4. Advantages of artificial neural network (ANN)

ANSWER:

- ❖ Excellent Pattern for recognition
- ❖ Excellent classifier
- ❖ Handles Noisy Data Well
- ❖ GOOD for generalization
- ❖ Efficiency.
- ❖ Continuous Learning.
- ❖ Hardware dependent.

5. differ between supervised and unsupervised learning.

ANSWER:

Supervised learning uses labeled datasets, whereas unsupervised learning uses unlabeled datasets. In a supervised learning model, input and output variables will be given. In unsupervised learning model, only input data will be given. Unsupervised learning is computationally complex. Supervised learning is a simpler method.

6. Mamdani fuzzy interface

ANSWER:

AL-JUNAID TECH INSTITUTE

The Mamdani fuzzy interface is a type of fuzzy logic system that uses linguistic rules to map inputs to outputs. It was developed by Lotfi Zadeh in the 1970s and is widely used in control systems and decision-making applications. The output of a Mamdani system is a fuzzy set that can be transformed into a crisp value using a defuzzification method.

7. Name any three soft computing algorithms

ANSWER:

- ❖ Fuzzy Logic
- ❖ Neural Networks.
- ❖ Genetic algorithms

8. Designs of artificial neural networks (ANN)?

ANSWER:

- ❖ Feature Representation
- ❖ Training
- ❖ Similarity measurement
- ❖ Validation
- ❖ Stopping Criteria
- ❖ Application Testing

9. What is Membership function? & its characteristics in fuzzy logic?

ANSWER:

A membership function is a mathematical function used in fuzzy logic to define the degree to which an element belongs to a fuzzy set.

Some characteristics of the membership functions are:

- ❖ It is represented by the Greek symbol μ
- ❖ Truth values range between 0.0 and 1.0
- ❖ Where 0.0 normally represents absolute falseness.
- ❖ And 1.0 represents absolute truth.

10. what is the CMD of CLIPS

ANSWER:

CMD (Command Prompt) is a command-line interface that can be used to interact with the CLIPS expert system shell. To open the CLIPS shell in CMD, you can first navigate to the directory where the CLIPS executable is located (using the 'cd' command), and then type 'clips' followed by the name of your CLIPS file (e.g., 'clips myexpertsystem.clp'). This will open the CLIPS shell and load your expert system file for execution.

AL-JUNAID TECH INSTITUTE

11. How we convert the And Or syntax convert into simple standard form

ANSWER:

- ❖ Distribute AND over OR using the distributive law:
 $a \text{ AND } (b \text{ OR } c) = (a \text{ AND } b) \text{ OR } (a \text{ AND } c)$
- ❖ Use De Morgan's laws to convert AND and OR between them:
 $\text{NOT } (a \text{ AND } b) = (\text{NOT } a) \text{ OR } (\text{NOT } b)$ and $\text{NOT } (a \text{ OR } b) = (\text{NOT } a) \text{ AND } (\text{NOT } b)$
- ❖ Simplify double negations:
 $\text{NOT } (\text{NOT } a) = a$

12. What is aggregation? Is it necessary for all the outputs of fuzzy inference system?

ANSWER:

Aggregation is the process by which the fuzzy sets that represent the outputs of each rule are combined into a single fuzzy set. The output of the aggregation process is one fuzzy set for each output variable. Without aggregation, the system would not be able to make a meaningful decision.

13. Illustrate minimax algorithm in AI?

ANSWER:

Minimax algorithm is a decision-making algorithm used in artificial intelligence to find the optimal move in a two-player game by minimizing the maximum possible loss.

14. Explain the Entropy?

ANSWER:

In artificial intelligence, entropy is often used as a measure of impurity in a decision tree. Entropy is a key component in many machine learning algorithms, such as decision trees, random forests, and information gain-based feature selection.

$$\text{Entropy}(S) = -p^+ \log_2 p^+ - p^- \log_2 p^-$$

15. What is hypothesis tree

ANSWER:

The hypothesis tree can help data scientists and AI researchers organize and structure their thinking about a particular problem, and can also help them

AL-JUNAID TECH INSTITUTE

identify the most promising paths for further exploration and experimentation.

16. Formula of fuzzy logic of AND OR NOT

ANSWER:

- ❖ Fuzzy AND: $\min(\mu_A(x), \mu_B(x))$
- ❖ Fuzzy OR: $\max(\mu_A(x), \mu_B(x))$
- ❖ Fuzzy NOT: $1 - \mu_A(x)$ where $\mu_A(x)$ represents the degree of membership of an element x in a fuzzy set A .

17. Printout product command in clips

ANSWER:

The Basic Syntax:

(printout-product [output-file] [pattern]);

For example, to display all rules and facts that have a pattern of (person (name JUNAID)), the following command can be used:

(printout-product stdout (person (name JUNAID)))

18. Explain Information gain?

ANSWER:

Entropy as a measure of the impurity in a collection of training. The effectiveness of an attribute in classifying the training data. The measure we will use, called information gain.

19. Five parts of fuzzy inference process?

ANSWER:

- ❖ Fuzzification of the input variable.
- ❖ Application of the fuzzy operator in the ancestor.
- ❖ Implication from antecedent to consequent.
- ❖ Aggregation of consequent across the rule.
- ❖ Defuzzification of output.

20. Conflict resolution strategies

ANSWER:

- ❖ Fire First in sequence
- ❖ Assign rules priorities
- ❖ More specific rule is preferred over the general rules
- ❖ Parallel Strategy

21. Briefly Explain hypothesis space

ANSWER:

AL-JUNAID TECH INSTITUTE

The collection of all potential legal hypotheses is known as hypothesis space. The machine learning system would choose the best candidate (and only one) from this set to best describe the goal function or the outputs.

22. Write the ESDLC steps?

ANSWER:

- ❖ Feasibility study
- ❖ Rapid prototyping
- ❖ Alpha system
- ❖ Beta system
- ❖ Maintenance and evolution

23. Valid data types used in CLIPS?

ANSWER:

- ❖ Numeric
- ❖ Symbol
- ❖ String

24. Write the Clips print command?

ANSWER:

(printout t "JUNAID MALIK" crlf)

25. Forward And Backward chaining

ANSWER:

SEE PAGE 123-126-128 (PDF HANDOUT)

26. CLIPS command to remove all data

ANSWER:

The Basic command is

CLIPS> (clear)

If we remove only one data from the given then

CLIPS> (retract 5)

27. Weak point of Find-S algorithm

ANSWER:

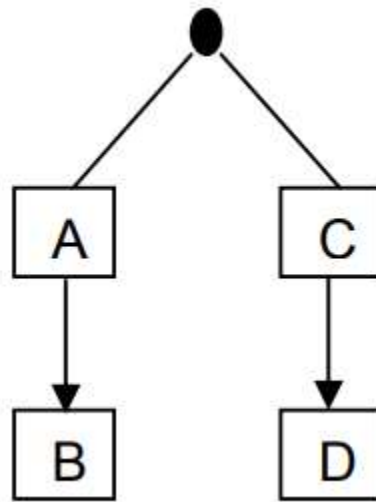
One weak point of the Find-S algorithm is that it is vulnerable to overfitting, where the learned hypothesis fits the training data too closely and may not generalize well to new, unseen data. Additionally, it can only learn conjunctions of attributes and may not be able to represent more complex relationships between attributes.

28. Decision tree representation of

(A AND B) OR (C AND D)

ANSWER:

AL-JUNAID TECH INSTITUTE



29. Clustering and name of any two-clustering algorithm

ANSWER:

Clustering is the process of grouping similar data points together based on their similarity or distance. The goal of clustering is to identify patterns or structures in data that can be used for further analysis or decision-making.

Two popular clustering algorithms are:

- ❖ Self-organizing maps (SOM)
- ❖ k-means
- ❖ Density based data analysis

30. In context of neural networks which term is used to ~~convert~~ weak signals into strong/large ones?

ANSWER:

Spatial summation occurs when several weak signals are converted into a single large one.

31. In context of hill climbing which problem is caused by multiple peaks in hill?

ANSWER:

A hill-climbing algorithm is a local search algorithm that moves continuously upward (increasing) until the best solution is attained. This algorithm comes to an end when the peak is reached.

32. Expert system phases?

ANSWER:

AL-JUNAID TECH INSTITUTE

- ❖ Identification
- ❖ Conceptualization
- ❖ Formalization
- ❖ Implementation

33.Clips facts program

ANSWER:

```
(printout t JUNAID MALIK crlf)
```

```
(deftemplate enrolled-courses
```

```
  (slot course-id)
```

```
  (slot course-name))
```

```
(assert (enrolled-courses (course-id "CS607") (course-name "Artificial  
Intelligence"))))
```

```
(assert (enrolled-courses (course-id "CS607P") (course-name "Artificial  
Intelligence Practical"))))
```

```
(facts)
```

34.Name of three type of problem.

ANSWER:

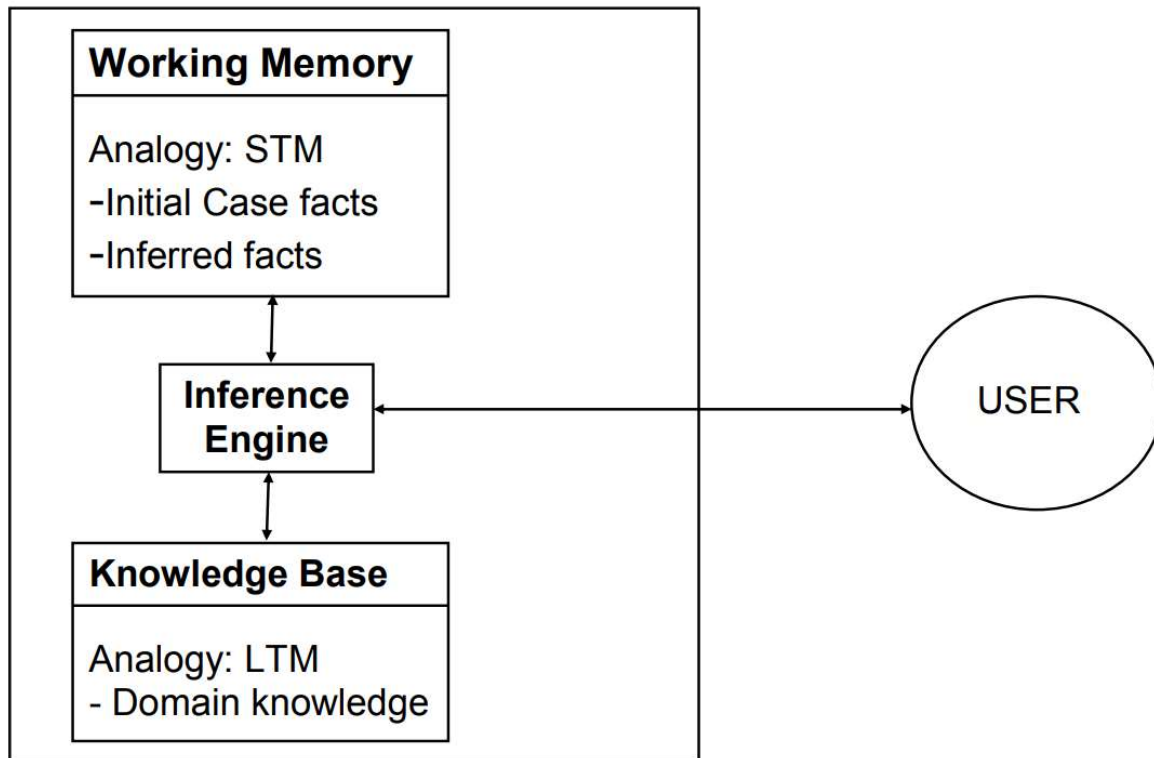
- ❖ Supervised learning
- ❖ Unsupervised learning
- ❖ Reinforcement learning

35.Expert system diagram

AL-JUNAID TECH INSTITUTE

ANSWER:

Expert System



36. Software engineering methodology are used for developing practical Expert Systems. You are required to write down the names of general stages of the expert system development lifecycle

ANSWER:

The general stages of the expert system development life-cycle or ESDLC are:

- ❖ Feasibility study.
- ❖ Rapid prototyping.
- ❖ Alpha system (in-house verification).
- ❖ Beta system (tested by users).
- ❖ Maintenance and evolution.

37. What is machine learning?

ANSWER:

The goal of machine learning is to build computer systems that can learn from their experience and adapt to their environments.

38. Write a command in CLIPS IDE to display “Virtual University”.

AL-JUNAID TECH INSTITUTE

ANSWER:

CLIPS> (printout t "Virtual University" crlf)

39. Convert the given statement into CNF: $(A \vee B) \rightarrow (C \rightarrow D)$.

ANSWER:

- ❖ $\sim (A \vee B) \vee (\sim C \vee D)$
- ❖ $(\sim A \wedge \sim B) \vee (\sim C \vee D)$
- ❖ $(\sim A \vee \sim C \vee D) \wedge (\sim B \vee \sim C \vee D)$

40. Which command is used to close the CLIPS program?

ANSWER:

CLIPS> (exit)

41. Write the main phases of linear sequence in development of expert system (ESDLC).

ANSWER:

- ❖ Planning
- ❖ Knowledge acquisition and analysis
- ❖ Knowledge design
- ❖ Code
- ❖ Knowledge verification
- ❖ System evaluation

42. In designing an expert system of car diagnostic system which basic step you have to follow. Write down all necessary steps.

ANSWER:

- ❖ Define the problem
- ❖ Define the input data
- ❖ Define the structure (Data driven)
- ❖ Write initial code
- ❖ Test
- ❖ Design the interface
- ❖ Expand the system
- ❖ Evaluate the system

43. Which topic in Artificial Intelligence deals with the concept of partial truth and membership function?

ANSWER:

Fuzzy logic is a topic in Artificial Intelligence that deals with and hence it has concept of partial truth and membership function.

AL-JUNAID TECH INSTITUTE

44.man(Ali) father(Ali, Fahad) Write down the commands to Add the above facts into CLIPS. Also write the command to modify the name "Ali" to "Ali Ahmed".

ANSWER:

```
CLIPS> (assert ( man ( name "Ali" ) ) )
```

```
CLIPS>(assert ( father ( fathersName "Ali") (sonsName "Fahad") ) )
```

```
CLIPS> (modify 2 ( fathersName "Ali Ahmed"))
```

45.What is the purpose of defuzzification? Name at least one method used for defuzzification.

ANSWER:

The process of defuzzification is conversion from Fuzzy to Crisp.

- ❖ One common method for defuzzification is called centroid. It uses the 'center of gravity' of the combined area under of memberships.

46.Define Computer vision in your own words

ANSWER:

Computer vision is a field of artificial intelligence that focuses on enabling machines to interpret and understand visual information from the world around us. It involves developing algorithms and models that can analyze, process, and make decisions based on images or videos. The ultimate goal is to create systems that can "see" and perceive the world like humans do, and use this understanding to solve a wide range of real-world problems.

47.Types of Knowledge acquisition

ANSWER:

- ❖ Manual knowledge acquisition
- ❖ Automatic knowledge acquisition
- ❖ Knowledge elicitation
- ❖ Knowledge transfer

48.Linearly separable problems and non-Linearly separable problems

ANSWER:

When two classes that can be separated by decision making. It is known as Linearly Problem. The simplest examples are the logical AND or OR.

When two classes cannot be separated by decision making line are known as non-Linearly separable. The main Example of Non-Linearly separable is XOR (exclusive OR)

AL-JUNAID TECH INSTITUTE

49. Write three drawbacks of ANN

ANSWER:

- ❖ Black box nature
- ❖ Need for large amounts of data
- ❖ Knowledge is implicit
- ❖ Data Dependency

50. Genetic Algorithms

ANSWER:

Genetic algorithms (GAs) are a type of optimization algorithm that are based on the principles of natural selection and genetics. They are commonly used to solve complex optimization problem.

Some advantages of genetic algorithms include their ability to handle large and complex search spaces, their ability to find multiple solutions, and their ability to perform well in noisy or uncertain environments.

51. Medical Imaging

NOTE: NOT PROPER QUESTION

ANSWER:

It analysis the different body scans like as X-Rays, CT scans, Bone Scans

52. Fuzzy logic

ANSWER:

Fuzzy logic is a type of logic that allows for degrees of truth or uncertainty, rather than strict binary values. It involves using fuzzy sets and membership functions to represent and reason about uncertain or vague concepts, and can be applied to a wide range of problems in decision-making, control, and pattern recognition.

53. Training and validation

ANSWER:

Training involves using a labeled dataset to train a machine learning model, while validation involves using a separate dataset to evaluate the performance of the model and tune its parameters.

54. Five steps of fuzzy logic

ANSWER:

- ❖ Fuzzification:
- ❖ Rule evaluation
- ❖ Aggregation
- ❖ Defuzzification
- ❖ Implementation

AL-JUNAID TECH INSTITUTE

55.What is Predicate action and it's Component.

ANSWER:

Predicate action is used to change the state it has three components.

- ❖ Predicate ITSELF
- ❖ Postcondition Predicate
- ❖ Preconditions Predicate

56.Candidate elimination algorithm

ANSWER:

The Candidate Elimination algorithm is a machine learning algorithm used for concept learning in the field of artificial intelligence. It is used to learn a hypothesis that fits the given data examples.

The Step of

- ❖ Initialize

The algorithm starts with two sets, S and G. S contains the most specific hypothesis that fits the first positive example, and G contains the most general hypothesis that fits the first negative example. Both sets are initialized to the same hypothesis, which is the most specific hypothesis possible (i.e., all attributes are set to 'null').

- ❖ For each training example

- a. If the example is positive, remove from S any hypothesis that is not consistent with the example.
- b. If the example is negative, remove from G any hypothesis that is consistent with the example.

- ❖ Output the hypothesis

- ❖ Repeat step 2

57.wetr vali example 0 to 10

NOTE: NOT PROPER QUESTION

ANSWER:

BUT YOU

SEE PAGE 154 (PDF HANDOUT)

58.Difference b/w inductive and deductive learning

ANSWER:

Inductive method is a method that is Move specific to example Generalization.

F.E:

AL-JUNAID TECH INSTITUTE

One child eats a green apple daily. The taste of green apple is sour. He observed in his daily routine Now the Green apple is specific. He found a conclusion all green apples are sour.

Deductive Learning is method that we move generalization to specific.

F.E:

We said all Green apples are sour.

Means in deductive method first we give the rule then test.

59. Five steps of expert System (ES)

ANSWER:

- ❖ Knowledge Gaining
- ❖ Inference Engine Development
- ❖ Testing and Validation
- ❖ Deployment and Maintenance

60. What is the purpose of fuzzification? Name at least one method used for fuzzification.

ANSWER:

The conversion of CRISP into a Fuzzy System. One method used for fuzzification is the membership function method.

61. What is STRIP?

ANSWER:

STRIP is one of the founding languages particularly for planning.

62. Write different knowledge Acquisition techniques.

ANSWER:

- ❖ On-site observation
- ❖ Electronic Brainstorming
- ❖ Brainstorming session with one or more Experts

63. How can you elaborate the condition predicate in STRIPS.

ANSWER:

The condition predicate in STRIPS is a logical expression that represents the preconditions required for an action to be executed.

64. Difference between fuzzy logic and Crisp value.

ANSWER:

Fuzzy logic is a type of Logic that allows the degree of truth or uncertain. It involves the fuzzy sets and member functions that represent and occasion about uncertain concept. It is applied to a wide range application for decision making.

The process of Converting the crisp input value to fuzzy.

AL-JUNAID TECH INSTITUTE

65. Who does neural network resemble the human brain.

ANSWER:

Neural networks resemble the human brain in their ability to process information through interconnected nodes that can adapt and learn.

66. How can you elaborate the importance and need of hypothesis space.

ANSWER:

The collection of a legal hypothesis is called a Hypothesis Space. The machine learning language would be choosing the best candidate from the set and best describe the goal functions.

67. Write down fuzzy statement in everyday life.

ANSWER:

Fuzzy Statement in daily life such as

- ❖ Engineering
- ❖ Manufacturing
- ❖ Metropology
- ❖ Medicine

68. Write down the task for which connectionist approach is well suited.

ANSWER:

- ❖ Classification
- ❖ Pattern Recognition
- ❖ Prediction

69. Why Genetic algorithm is used in the biological theory of Human evolution? Also write its functions.

ANSWER:

Genetic algorithm can inspire by a biological process of a natural selection and evolution.

Functions:

- ❖ Initialization
- ❖ evaluation
- ❖ Selection
- ❖ Mutation
- ❖ Crossover
- ❖ Termination

70. What is Mean Squared error?

ANSWER:

The most common technique for measuring the total error in each iteration of the neural network (epoch) is Mean Squared Error (MSE).

AL-JUNAID TECH INSTITUTE

71.steps of back ward chaining

ANSWER:

- ❖ Start with goal
- ❖ Goal may be in WM initially

72.what is the difference between knowledge analysis and evaluation function

ANSWER:

Knowledge analysis is a process of transforming raw data into usable knowledge, while an evaluation function is a mathematical function used to evaluate the quality of a given move or state in decision-making algorithms.

73.Fuzzy inference system (FIS) Application.

ANSWER:

FIS is a type of expert system that use fuzzy logic to make decision.

74.conjunctive normal form conversion.

ANSWER:

Resolution requires all sentences to be converted into a special form called conjunctive normal form (CNF). A statement in conjunctive normal form (CNF) consists of ANDs of Ors.

75.What is the prime role of expert system designer?

ANSWER:

The prime role of an expert system designer is to develop a knowledge-based system that can effectively solve complex problems in a specific domain by incorporating the knowledge and expertise of human experts.

76.Among Entropy and information gain which is the best to measure effectiveness?

ANSWER:

Information gain is the best measure effectiveness.

77.Why clustering algorithm are categorized as unsupervised?

ANSWER:

Clustering algorithms are categorized as unsupervised because they do not require labeled data or predefined classes to group similar data points together.

78.Which term is best suited for a function that is used specially to drive out the degree of truth in fuzzy sets? Write its two characteristics.

ANSWER:

AL-JUNAID TECH INSTITUTE

A membership function is a mathematical function used in fuzzy logic to define the degree to which an element belongs to a fuzzy set.

Some characteristics of the membership functions are:

- ❖ It is represented by the Greek symbol μ
- ❖ Truth values range between 0.0 and 1.0
- ❖ Where 0.0 normally represents absolute falseness.
- ❖ And 1.0 represents absolute truth.

79. Genetic Algorithm is inspired by the structure and/or functional aspects of the biological neural networks and it consist of an interconnected group of artificial neurons. Do you agree or not?

ANSWER:

No, Genetic Algorithm is not inspired by the structure or functional aspect BUT it can be inspired by the biological of natural selection and evaluation.

80. How expert system work

ANSWER:

A knowledge expert person gives all the knowledge to the expert system. This is known as knowledge base and It has as interface engine. Interface engine contain all facts and related data can store. When user send his query to expert system the interface engine read the query and check the related data and give a solution. Interface engine is the brain of Expert System.

81. Which measurement issued as total error for the stopping criteria in neural network for epochs

ANSWER:

The most commonly used measurement for the stopping criteria in neural network training is validation loss.

82. In softcomputing the most combinations are of

I) genetic -fuzzy system

II) n eural-fuzzy system

III) genetic algorithm -neural network

the question was identify and describe which combination is best suited

ANSWER:

- ❖ genetic algorithms – fuzzy systems (genetic fuzzy)
- ❖ Neural Networks – fuzzy systems (neuro-fuzzy systems)
- ❖ Genetic algorithms – Neural Networks (neuro-genetic systems)

AL-JUNAID TECH INSTITUTE

83.Information gain is a measure of the impurity in a collection of training examples. Do you agree with the statement or not? If not, then write a correct statement.

ANSWER:

No, I don't agree with this statement

The correct statement is

ENTROPY is used for the measure of IMPURITY to decision tree.

84.Which term is best suited for a situation in which we perform unsupervised learning on training data without having classification information or class labels?

ANSWER:

CLUSTERING

85.Draw the truth table (for two operands A and B) for Fuzzy operator AND.

ANSWER:

A	B	A and B
0	0	0
0	1	0
1	0	0
1	1	1

AND

86.Draw the truth table (for two operands A and B) for Fuzzy operator OR.

ANSWER:

AL-JUNAID TECH INSTITUTE

A	B	A or B
0	0	0
0	1	1
1	0	1
1	1	1

OR

87. Draw the truth table (for two operands A and B) for Fuzzy operator NOT.

ANSWER:

A	not A
0	1
1	0

NOT

88. ANN strategy which uses training data

ANSWER:

The ANN strategy that uses training data is to adjust the network's parameters

89. CRISP is BOOLEAN two-valued function is statement correct. Support answer

ANSWER:

No, this is statement is incorrect. CRISP is not a function

CRISP is an acronym that stands for the steps involved in the process:

❖ Business Understanding

AL-JUNAID TECH INSTITUTE

- ❖ Data Understanding
- ❖ Data Preparation
- ❖ Modeling
- ❖ Evaluation
- ❖ Deployment

90.Elicitation methods category

ANSWER:

- ❖ Indirect Method
- ❖ Direct Method

91.Write Plaining Phase.

ANSWER:

- ❖ Feasibility assessment
- ❖ Resource Allocation
- ❖ Requirement Analysis

92.Write the primary composed thing.

ANSWER:

- ❖ Input
- ❖ Processing unit
- ❖ Output

93.Write the branch of Theoretical computer.

ANSWER:

- ❖ Tractable
- ❖ Intractable

0304-1659294

www.vulmshelp.com